

William Greenfield

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LinkedIn GitHub

Profile

A motivated and analytical computer science student entering his final year of a BSc at the University of Southampton, with strong academic ability, practical experience in software engineering, machine learning and data analysis, and the tenacity to see things through. An enthusiastic problem solver with good coding, verbal and written communication skills. Developed the ability to define open-ended research problems, evaluate incomplete evidence, and use programming to determine a practical route forward when no fixed specification exists. A self-starter who works well as part of a team, whilst also having the ability to lead and make decisions.

Experience

University of Southampton

Undergraduate Research Intern

Southampton, UK

Jun–Aug 2026

- Conducted an interdisciplinary literature review on biological synapses, memristive devices, and their biological applications under the supervision of Dr Liudi Jiang and Dr Ruomeng Huang.
- Compiled data from multiple experimentally measured datasets and published literature to construct an end-to-end semi-synthetic trace where measured data were scarce, using Python, pandas, and NumPy.
- Evaluated existing software models and simulated novel memristive devices as artificial synapses
- Integrated the artificial-synapse models with a published neuron model to simulate signal transfer through a neuron-synapse pathway.
- Developed supporting software, including interactive web animations and a graphing platform for comparing waveforms and viewing benchmark data.

Ibex.ai - Software & Data Engineering Intern

AI and data-driven marketing platform

London, October 2022

- Queried production databases and used data pipelines to extract and clean a customer dataset using Pandas, enabling downstream ML analysis.
- Trained and evaluated regression and classification models on Azure Databricks to model the correlation between marketing strategy and client response rates.

IoT.nxt - Data Analysis Intern

IoT solutions provider

Gauteng, South Africa, July 2022

- Extracted match performance statistics of provincial rugby players and analysed this data, preparing a report with findings.

Codehesion - Software Development Intern

Software development consultancy

Gauteng, South Africa, July 2022

- Supported a team developing a supply chain management system for a large high street retailer.
- Developing customer loyalty system for online gaming company.

Selected Projects

Final-Year Dissertation

Adversarial Defence for Robust Evidence-Based Stance Detection

University of Southampton

2026–2027

- Researching natural language processing methods that classify whether text supports, opposes, or is neutral towards a claim.
- Developing adversarial defences that make stance predictions more reliable and better grounded in evidence from the source text.

Neural Networks Research Project

- Self-directed extended project researching the mathematics underpinning neural networks; built a feed-forward network from scratch in Python using only NumPy for matrix operations, achieving 99% accuracy on the MNIST handwritten-digit dataset.
- Investigated the effect of activation functions, learning rate and network architecture on training convergence and accuracy.

AI Coursework: 5×5 Chess Engine

- Built a 5×5 chess engine in Python using bitboards and Zobrist hashing for fast move generation, position tracking and transposition detection.
- Implemented iterative-deepening minimax with alpha-beta pruning, quiescence search and move-ordering heuristics, allowing the engine to search significantly deeper within a fixed time budget.
- Later I ported the engine to JS so it can run on my website

Maze Solving Program

- Led a team to design and implement a maze solving program using graph theory to navigate and solve conventional and unconventional mazes.
- Used Python cv2 library for image processing, A* search, breadth-first search (BFS) and depth-first search algorithms (DFS) to search the webcam input.

Web-Based Chess Platform

- Developed a web based two player chess game, using Flask and WebSockets interacting with the backend which provided user accounts, move validation utilising an Object-Oriented implementation of pieces.
- To allow for ease of use the web interface was built with similar features to popular existing chess platforms.

Education

University of Southampton

Southampton, UK

BSc Computer Science

2024–2027

Year 2 average: 65.1% (Upper Second). Year 1 average: 70%.
Selected marks: Artificial Intelligence 86%; Programming Language Concepts 72%; Formal Specification and Verification 71%; Programming I 77%; Programming II 77%; Algorithmics 73%.

Hampton School

London, UK

A Levels

2024

Mathematics A*, Further Mathematics A, Computer Science A, Physics A.

Technical Skills

Programming	Python, Java, JavaScript, TypeScript, C, Haskell, SQL
Machine learning and data	PyTorch, NumPy, Pandas, TensorFlow
Software and web	Git, Linux, Bash, Flask, React, HTML/CSS, WebSockets

Awards and Activities

- **1st Place, University of Southampton Cyber Security Society CTF**, May 2025. Ranked first across cryptography, web security, reverse engineering, binary exploitation, RFID hacking, and protocol engineering.
- Founded and led Hampton School’s Competitive Programming Club. Served as School Prefect in 2023–2024.

Volunteering

Nkosi’s Haven

Johannesburg, South Africa

Volunteer, shelter for mothers and children living with HIV

Aug 2022

- Supported the day-care centre, organised after-school activities, and helped with kitchen work and meal preparation.